

BULMAN, AustraliaWMO: **941430**Lat: **13.6714S**Lon: **134.3414E**Elev: **105**StdP: **100.07**Time Zone: **9.50 (AUC)**Period: **04-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	11.8	1.1	4.2	27.0	3.0	4.8	26.7	6.4	27.7	6.1	27.9	0.3	200	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	13.1	39.0	22.2	38.1	22.3	37.3	22.3	26.7	32.4	26.2	31.8	25.9	31.5	3.3	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	20.7	28.2	25.0	20.3	28.0	24.6	19.9	27.8	84.3	32.5	82.4	32.0	81.0	31.6	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
6.1	5.4	5.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			7.5	40.5	2.1	0.6	5.9	40.9	4.7	41.3	3.4	41.6	1.9	42.0
			6.5	28.1	2.3	1.8	4.9	29.4	3.5	30.5	2.2	31.5	0.6	32.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	29.3	29.0	28.5	27.0	25.1	22.5	22.3	23.4	26.6	29.3	30.8	30.5	
	DBStd	3.56	1.36	1.44	1.35	1.64	2.33	2.92	2.55	2.38	2.26	1.90	1.51	1.66	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	7	0	0	0	0	0	3	3	1	0	0	0	0	
	CDD10.0	6213	597	533	574	510	467	376	383	414	497	599	625	637	
	CDD18.3	3178	339	300	316	260	209	129	127	157	247	341	375	378	
	CDH23.3	38698	3888	3376	3401	2771	2126	1264	1413	2048	3401	4867	5221	4924	
	CDH26.7	19967	1830	1607	1612	1324	904	440	524	955	1943	2958	3133	2737	
Wind	WSAvg	2.1	2.0	1.8	1.6	1.8	2.1	2.3	2.3	2.3	2.3	2.4	2.3	2.0	
Precipitation	PrecAvg	1207	266	273	211	67	15	7	5	5	9	30	96	222	
	PrecMax	1839	436	504	481	301	53	23	23	25	89	146	213	350	
	PrecMin	810	109	110	40	3	2	2	1	1	1	1	32	88	
	PrecStd	278	96	104	118	66	13	5	4	4	16	33	39	80	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.9	37.5	37.2	35.7	34.4	33.2	33.1	35.1	37.5	39.4	40.2	40.2	
		MCWB	24.3	24.2	23.9	22.8	22.6	20.9	20.5	20.1	20.4	21.5	22.4	22.0	
	2%	DB	36.4	36.2	35.6	34.7	33.2	31.9	31.9	33.7	36.5	38.3	39.1	38.8	
		MCWB	24.7	24.4	24.1	22.7	22.1	20.8	19.9	19.6	20.0	21.4	22.1	22.9	
	5%	DB	35.1	35.2	34.6	33.9	32.2	30.6	30.9	32.6	35.7	37.5	38.1	37.6	
		MCWB	25.3	24.9	24.2	22.9	21.8	20.2	19.3	19.0	19.8	21.2	22.1	23.5	
	10%	DB	33.8	34.0	33.5	32.8	31.2	29.4	29.8	31.6	34.7	36.6	37.2	36.3	
		MCWB	25.3	25.1	24.5	22.8	21.5	19.6	18.7	18.6	19.5	21.0	22.2	23.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	27.2	27.0	25.8	25.4	23.1	23.0	22.5	24.0	24.7	25.9	26.7
			MCDB	33.3	32.2	32.6	30.6	30.6	28.5	29.2	30.2	29.0	31.7	33.1	33.0
2%		WB	26.6	26.6	26.4	25.5	24.5	22.3	22.0	21.6	23.2	24.1	25.2	26.1	
		MCDB	32.6	32.1	31.6	30.1	29.5	27.8	28.0	29.4	29.5	31.0	32.0	32.2	
5%		WB	26.1	26.1	26.0	25.0	23.9	21.7	21.2	20.9	22.5	23.6	24.7	25.6	
		MCDB	31.9	31.9	31.0	29.5	28.4	27.3	27.3	28.7	29.7	30.8	31.4	31.6	
10%		WB	25.8	25.7	25.6	24.5	23.3	21.0	20.3	20.1	21.9	23.2	24.3	25.3	
		MCDB	31.2	31.2	30.3	28.8	27.8	26.9	26.5	28.3	29.8	30.7	31.1	30.9	
Mean Daily Temperature Range	5% DB	MDBR	9.3	9.6	10.1	11.9	12.8	13.7	15.2	16.8	16.8	15.0	13.1	11.2	
		MCDBR	11.4	11.7	12.3	13.6	13.6	13.8	15.7	17.4	17.6	15.6	14.6	13.5	
		MCWBR	3.0	3.0	3.2	4.2	4.4	4.7	5.3	5.5	4.9	3.8	3.3	3.0	
	5% WB	MCDBR	9.3	9.5	9.9	10.0	10.6	11.8	12.8	15.0	13.5	13.4	12.2	10.7	
		MCWBR	3.0	3.2	3.1	3.2	3.6	4.1	4.7	5.0	3.6	3.3	2.9	2.7	
Clear-Sky Solar Irradiance	taub	0.404	0.399	0.392	0.374	0.362	0.340	0.335	0.340	0.395	0.417	0.432	0.416		
	taud	2.504	2.518	2.532	2.559	2.556	2.574	2.578	2.535	2.370	2.336	2.346	2.447		
	Ebn at Noon	940	937	925	909	887	893	907	929	908	910	908	927		
	Edh at Noon	115	113	108	100	94	89	91	100	125	134	134	121		
All-Sky Solar Radiation	RadAvg	5.88	5.89	5.72	5.61	5.17	4.94	5.27	6.06	6.75	7.05	6.79	6.37		
	RadStd	0.52	0.70	0.62	0.58	0.34	0.28	0.21	0.22	0.35	0.40	0.27	0.46		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.40	N/A	N/A	N/A	N/A	-0.47	N/A	N/A	N/A	N/A

Nomenclature: See separate page